Visualizing Waypoints-Constrained Origin-Destination Patterns for Massive Transportation Data

Computer Graphics Forum (CGF)

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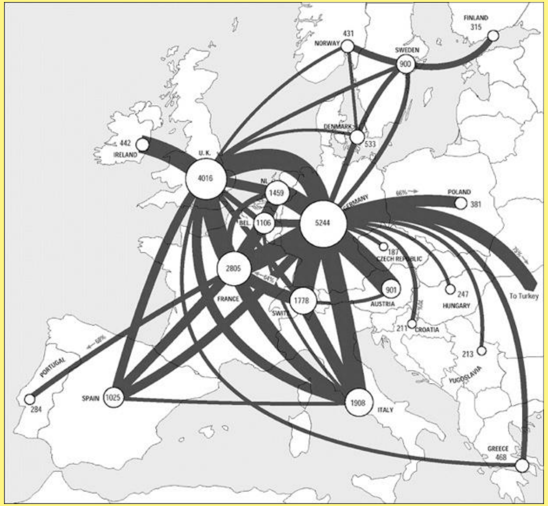
1.1 Origin-Destination (OD) Pattern

- OD pattern summarizes the number of trips going from each origin to each destination.

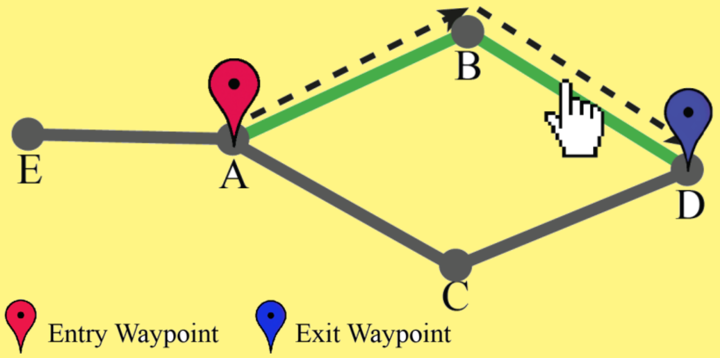
People Flow During Chinese New Year



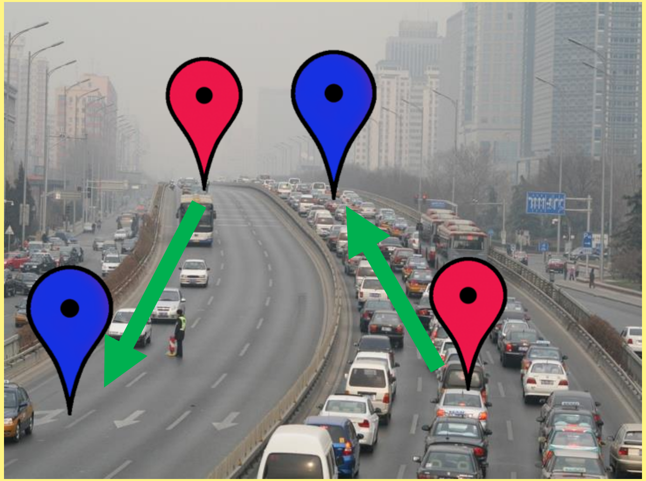
Traffic Flows Across Europe

1.2 Waypoints-Constrained OD Pattern

- Associates with trips *successively* pass through an entry and exit waypoints in an urban transportation network.



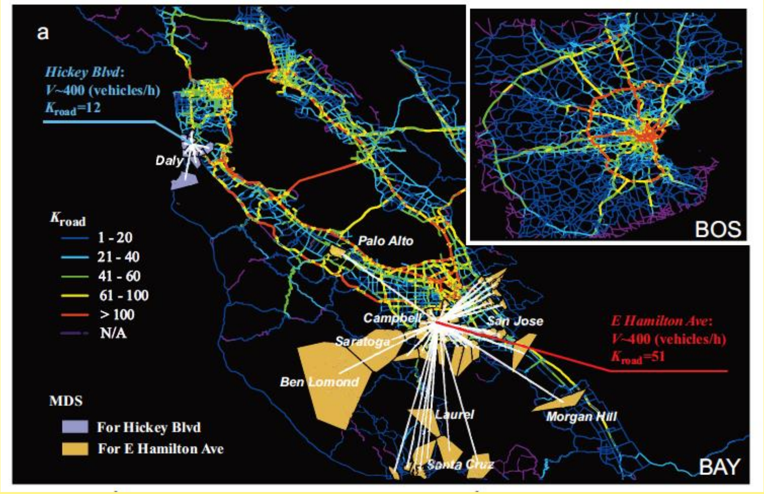
Waypoints illustration



Real world scenario: traffic jam in one direction, but not the other

1.3 Motivation

- Study shows that only a few (less than 2%) of the road segments in urban areas give rise to congestion, and the number of origins and destinations are limited [Wang et al., 2012].



Road usage patterns (Wang et al., 2012)

Transportation analysis scenario

1.4 Analytical Tasks

- **Query**: support interactive filtering of trajectories, say *{T}.*
- **Visual representation**: spatial and temporal information:
  - Task 1: origins and destinations from {T};
  - Task 2: flow volumes among the OD pairs derived from {T};
  - Task 3: temporal changes in flow volumes among the OD pairs.
- **Visual representation:** path-related task:
  - Task 4: paths through which the trajectories go from origins to destinations.

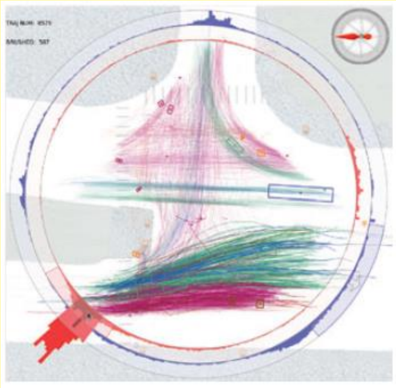
1.5 Visualization Challenges

- **Many OD pairs**: directly connect all OD pairs can easily cause visual clutter.
- **Vast amount of trips:** efficiently filter out relevant trips from millions of trips per day.
- **Temporal variations:** compare peak versus non-peak hours.
- **Waypoints constrained:** specify entry and exit waypoints
- **Movement paths:** present the paths of trips from origins to destinations, and examine multiple paths in-between.

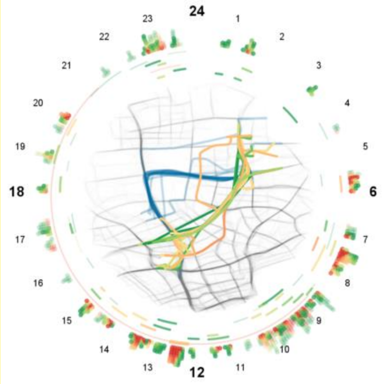
Contents

2.1 Visual Analytics of Movements

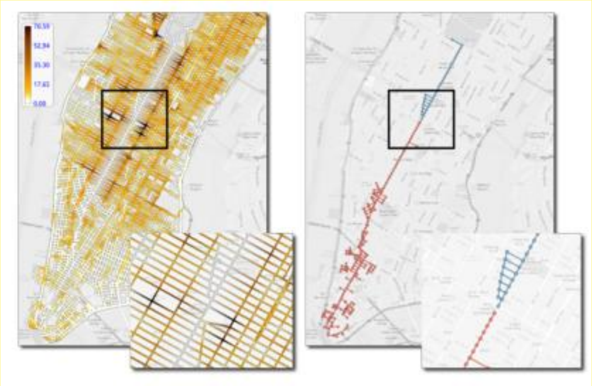
- Movements: a set of discrete objects whose positions change over time [Hornsby & Egenhofer, 2002].
- Urban traffic visual analytics
  - Facilitate the understanding of urban dynamics and human activities
  - Enhance traffic management and transportation planning



[Guo et al., 2011]



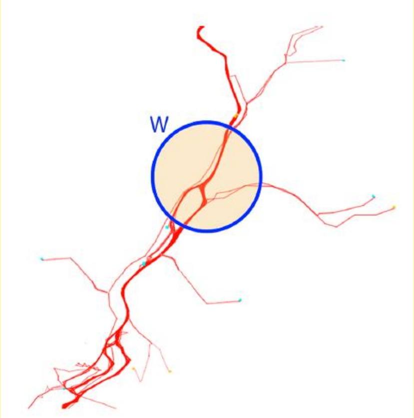
[Liu et al., 2011]



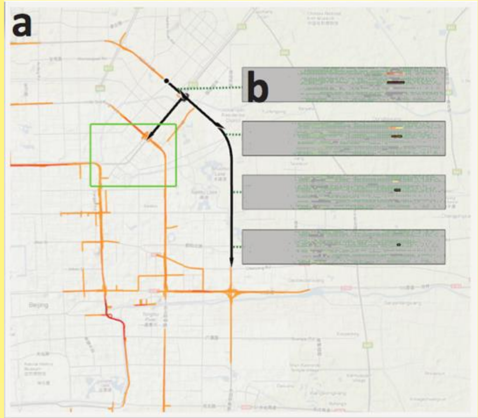
[Doraiswamy et al., 2014]

2.1 Visual Analytics of Movements

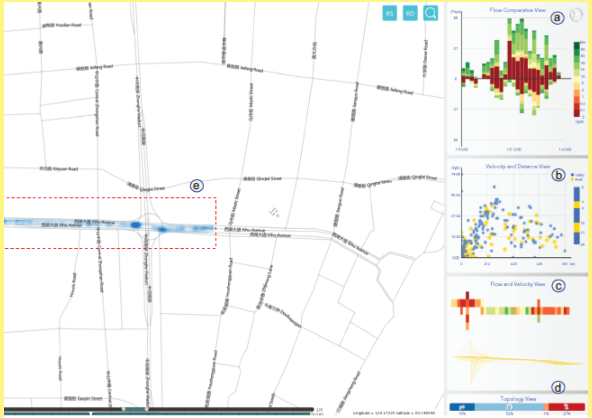
- Many visual analytics have also been developed to study movements along specific urban roads.
- None of them focuses on analyzing OD patterns.



Waypoint lens
[Krüger et al., 2013]



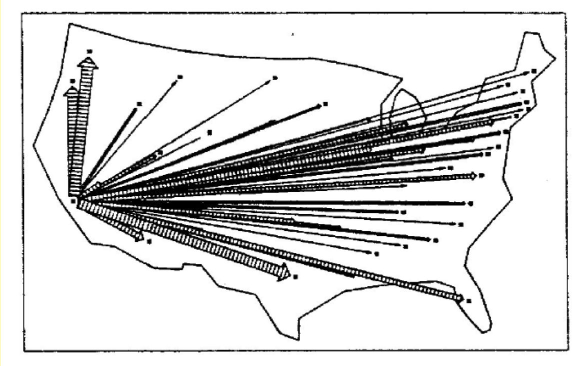
Traffic jam analysis
[Wang et al., 2013]



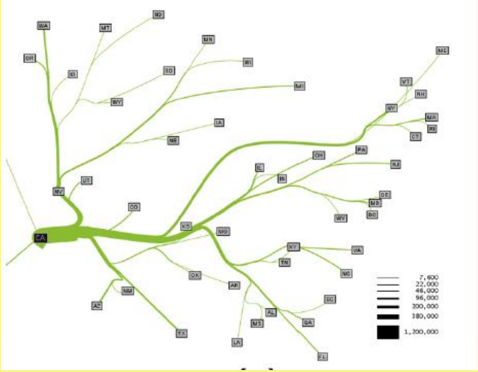
Transport assessment on urban roads
[Wang et al., 2014]

2.2 OD Visualization: Flow Map

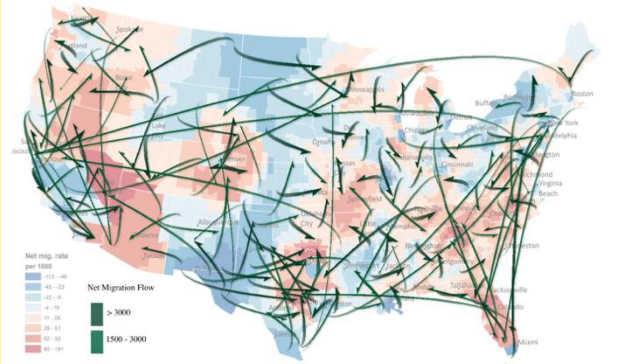
- Flow map joins origins and destinations by straight/curved arrows with line width indicating aggregated flow volume.
- Typical flow map remains the challenges of visual clutter and occlusion, MAUP, and normalization.



Tobler’s flow map
[Tobler, 1987]



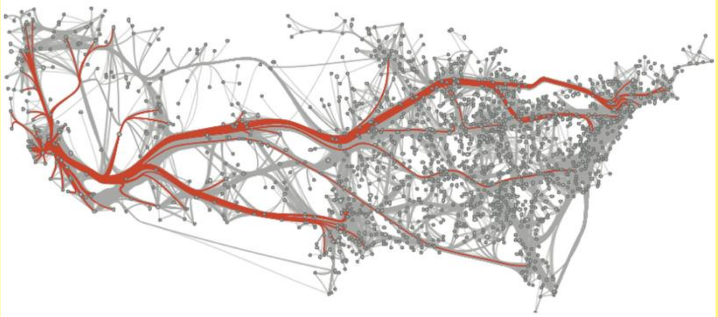
Flow map layout
[Phan et al., 2005]



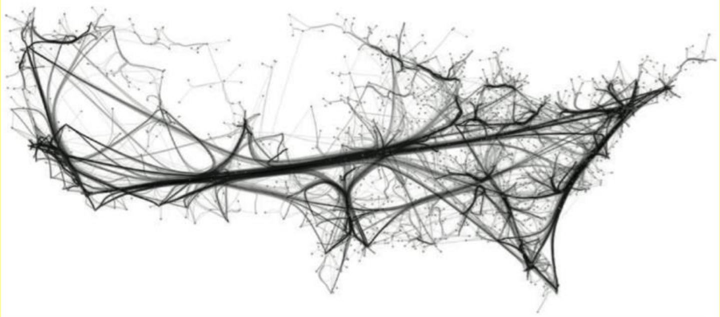
Smoothing flow map
[Guo and Zhu, 2014]

2.2 OD Visualization: Node-Link Graph

- Node-link graph represents origins and destinations as nodes, and flows in-between as directed links.
- Edge bundling can improve the readability, but issues like flow directions and overlapping remain challenging to address.



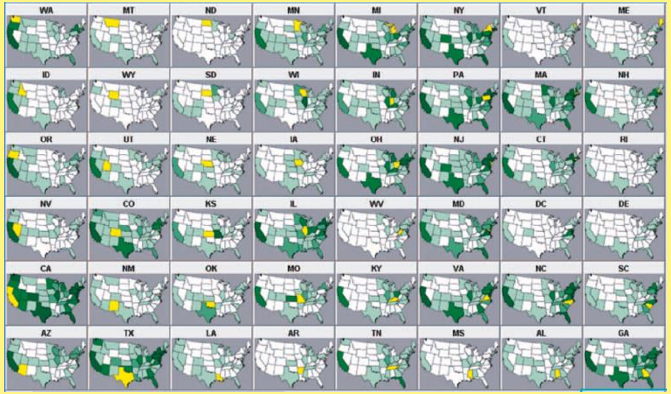
Geometry-based edge bundling
[Cui et al., 2008]



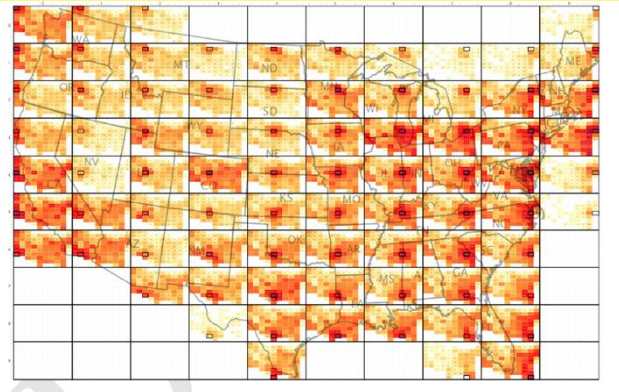
Force-directed edge bundling
[Holten and van Wijk, 2009]

2.2 OD Visualization: Matrix-based

- Matrix-based OD visualization summarizes the movements as an OD matrix, and each cell is symbolized with a color representing the magnitude of flow.
- Temporal and movement paths are hard to be integrated.



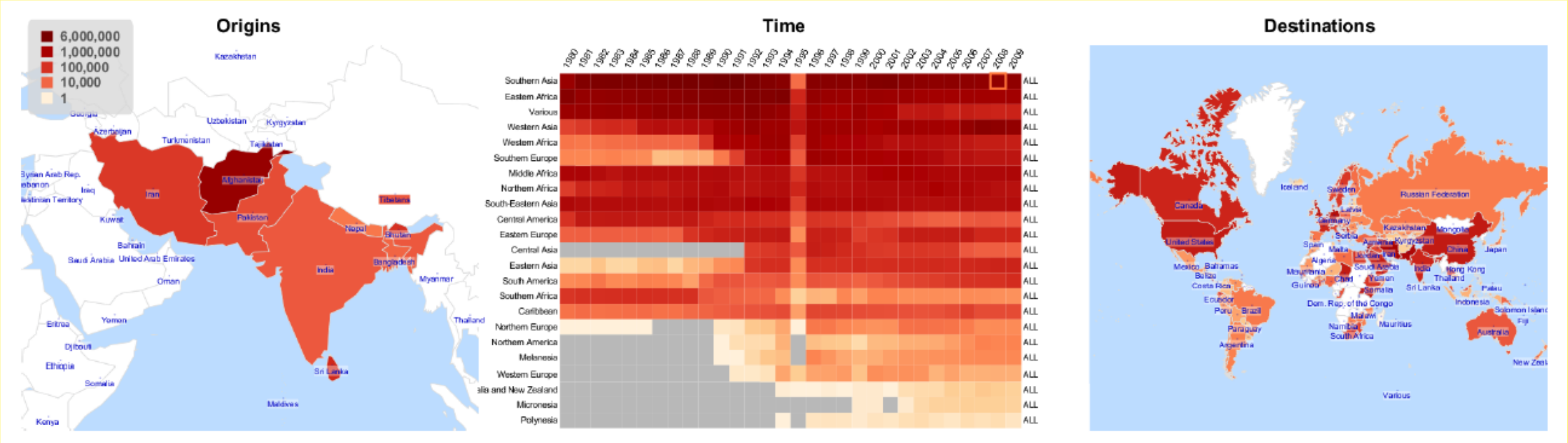
VIS-STAMP
[Guo et al., 2006]



OD Map
[Wood et al., 2010]

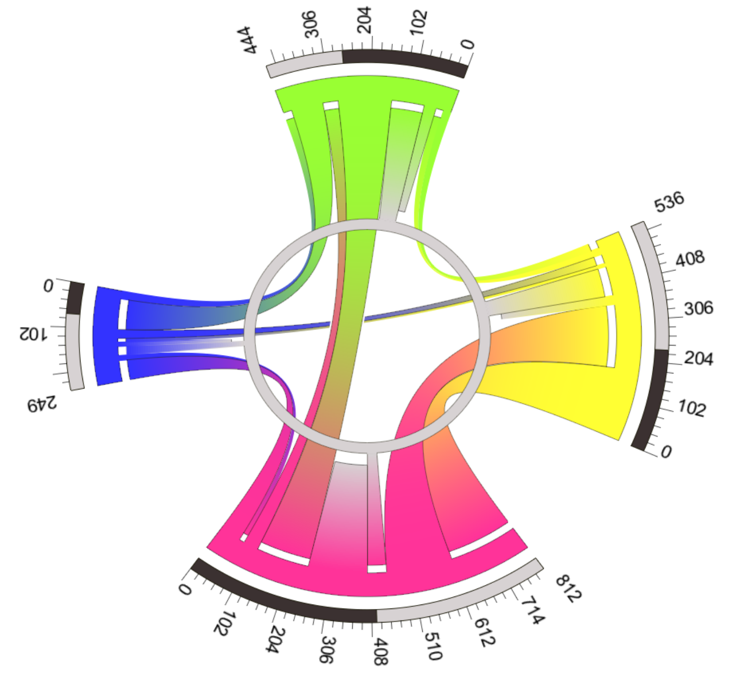
2.2 OD Visualization: Temporal OD Data

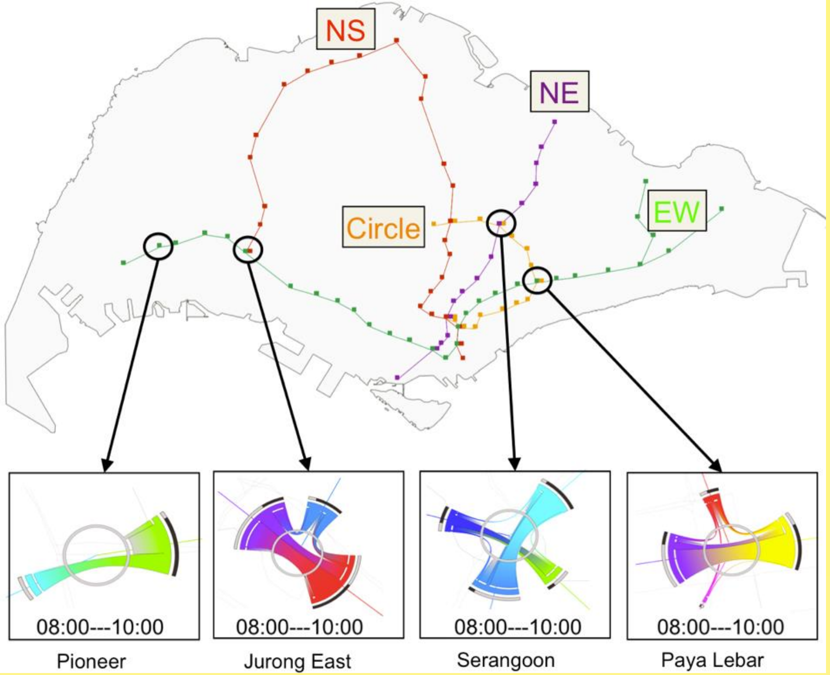
- Small multiples are not scalable, and animations are difficult to explore changes.
- Flowstrates shows ODs on two separate maps on the left and right, and presents temporal changes by a heat map in the middle [Boyandin et al., 2011].



2.3 Our Previous Work

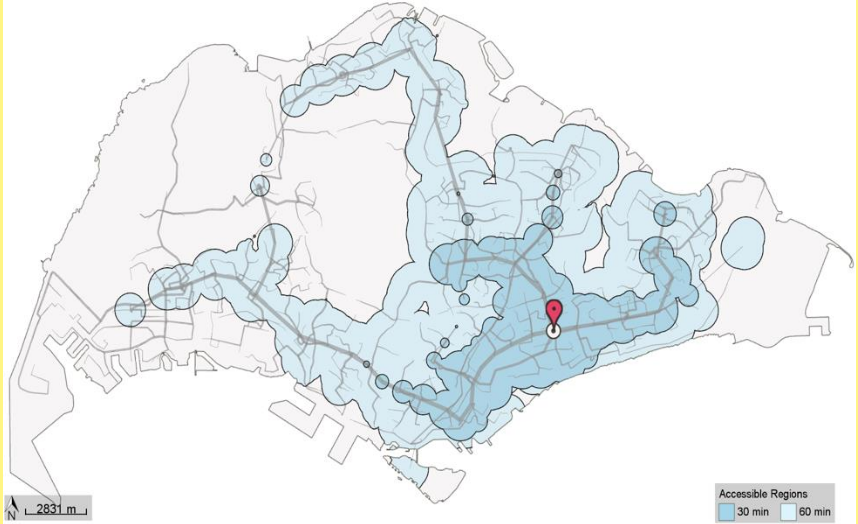
- Local OD visualization: Interchange Circos Diagram

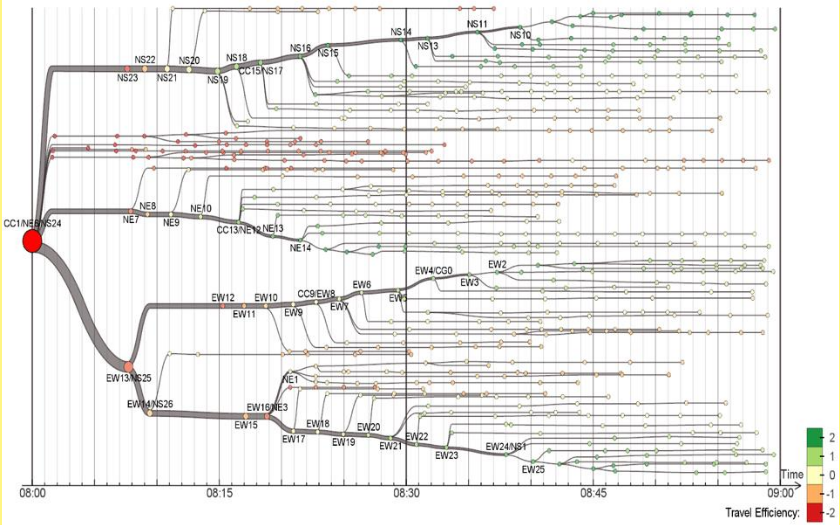




2.3 Our Previous Work

- 1-to-N OD visualization: Mobility Visualization





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Contents

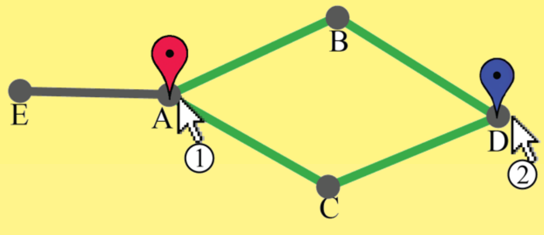
3.1 Data and Preprocessing

- 4 Mass Rapid Transit (MRT) lines, and 88 MRT stations in Singapore in 2011.
- ~2.1 million MRT trips are recorded daily.
  - **Boarding Stop:** Changi Airport
  - **Alighting Stop:** Jurong East
  - **Ride Start Time:** 09:01:03
  - **Ride Time:** 43.45 min
- Trip routes are reconstructed in preprocessing stage [Erath et al., 2012].

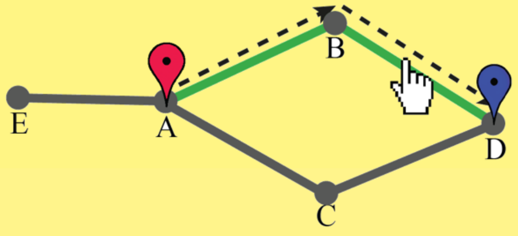
Singapore public transportation network in 2011

3.2 Waypoint-Constrained OD Query

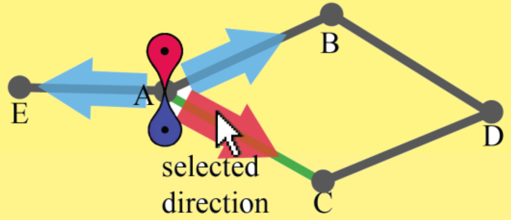
- Interactive waypoints specification:
  - Mode 1: two successive clicks
  - Mode 2: drag along a path
  - Mode 3: click to select an entry waypoint and an outgoing direction
  - Modify a waypoint through dragging
- Hashing-based trip indexing scheme



Mode 1



Mode 2

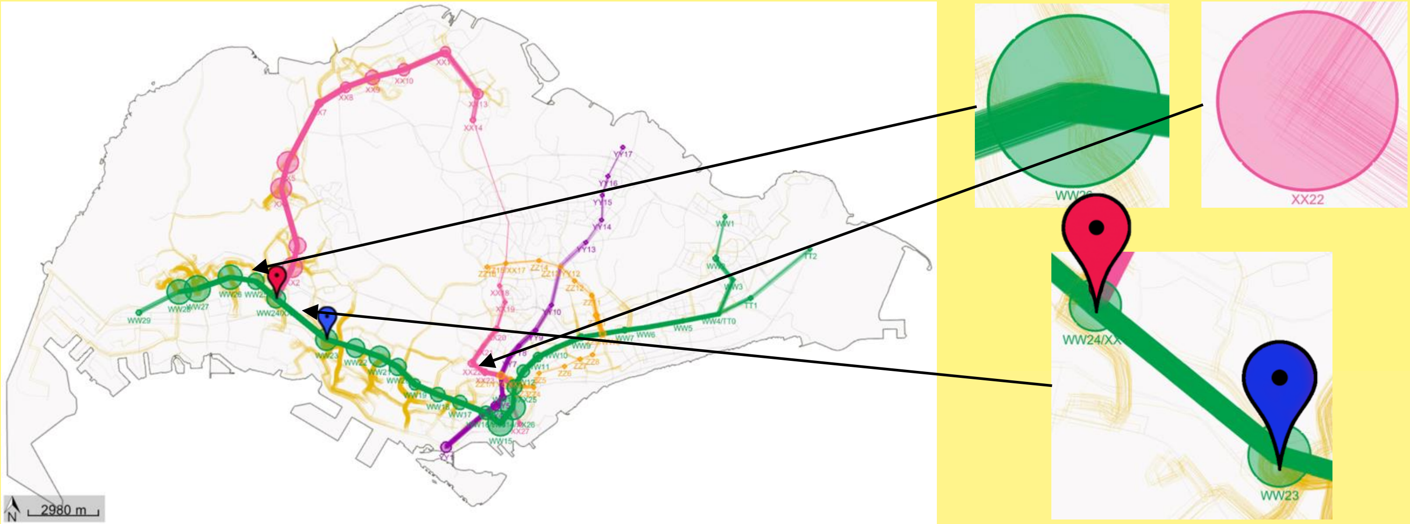


Mode 3

3.3 Map View

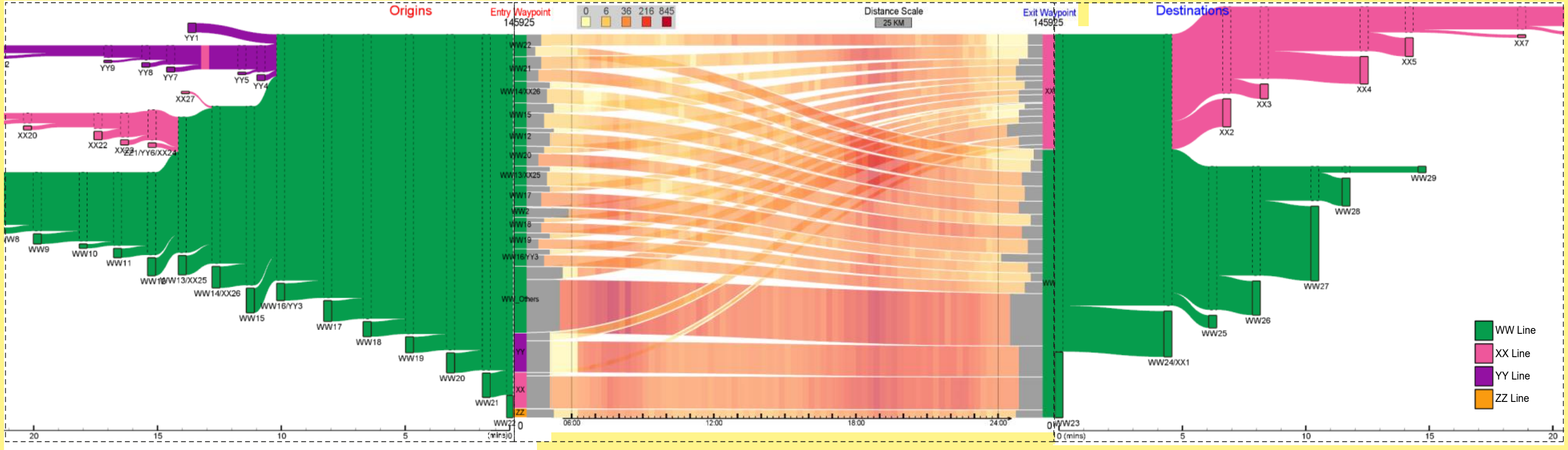
➢ Overview the filtered trips on the map for locating the origins and destinations in the physical space (Task 1)

- Origins and destinations: hollow circles
- Entry and exit waypoints: red and blue glyphs
- Filtered trajectories: randomly jitter the position of trajectory paths



3.4 Waypoints-Constrained OD View

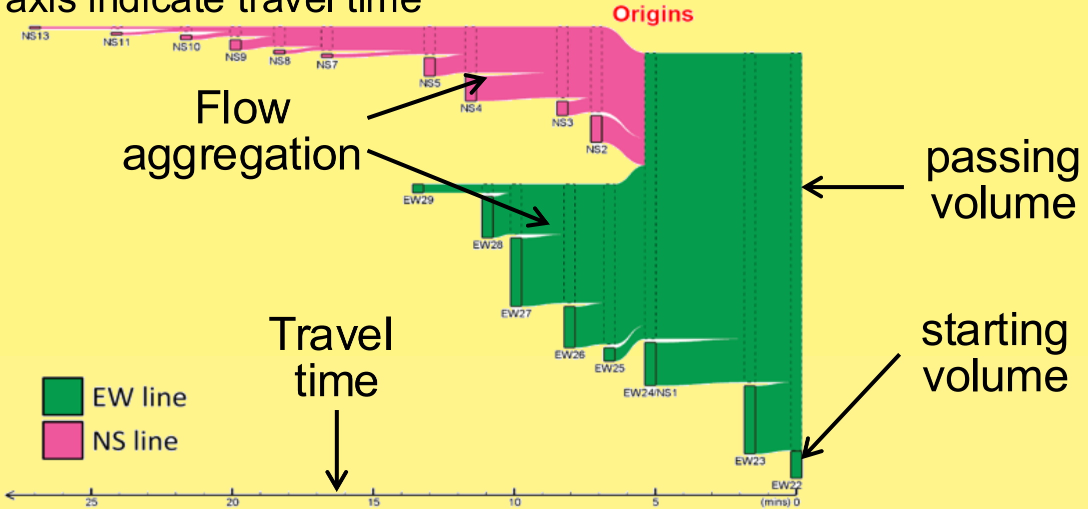
- Integrated three components to support the tasks:
  - In-flow view
  - OD-flow temporal view
  - Out-flow view



(a) In-Flow View (b) OD-Flow Temporal View (c) Out-Flow View

3.4.1 In- and Out-Flow View

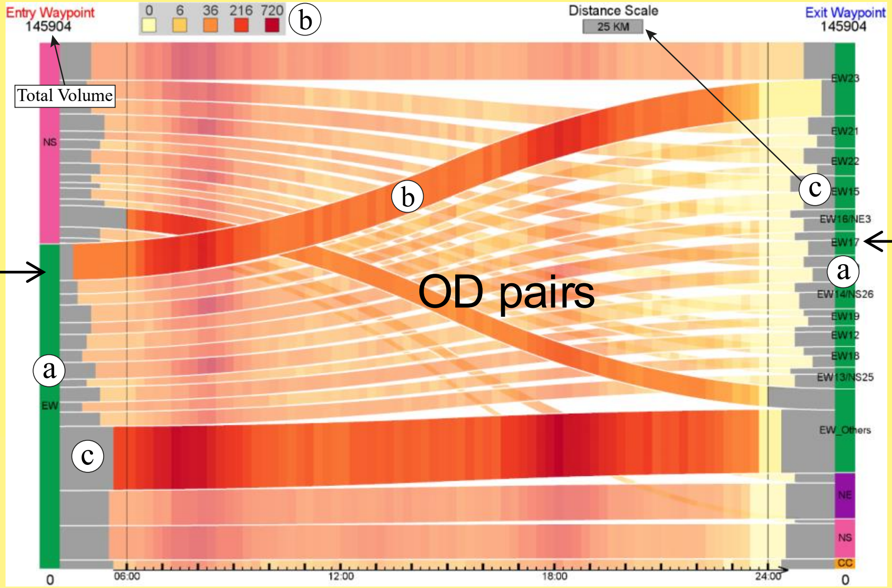
- Compare flow volumes of different stations (*Task 1*)
  - Solid boxes represent starting (ending) volumes
  - Dashed boxes represent passing volumes
- Visualize the paths from origins to destinations (*Task 4*)
  - Flows aggregate (distribute) from left to right
  - Horizontal axis indicate travel time



3.4.2 OD-Flow Temporal View

- Sankey diagram integrated with heat map to show the OD-pair flow volumes and temporal variations (*Tasks 2 & 3*).
  - (a) Vertical bars: origins (left) and destinations (right)
  - (b) Ribbons: OD-pair flows and their temporal variations
  - (c) Relative travel distance

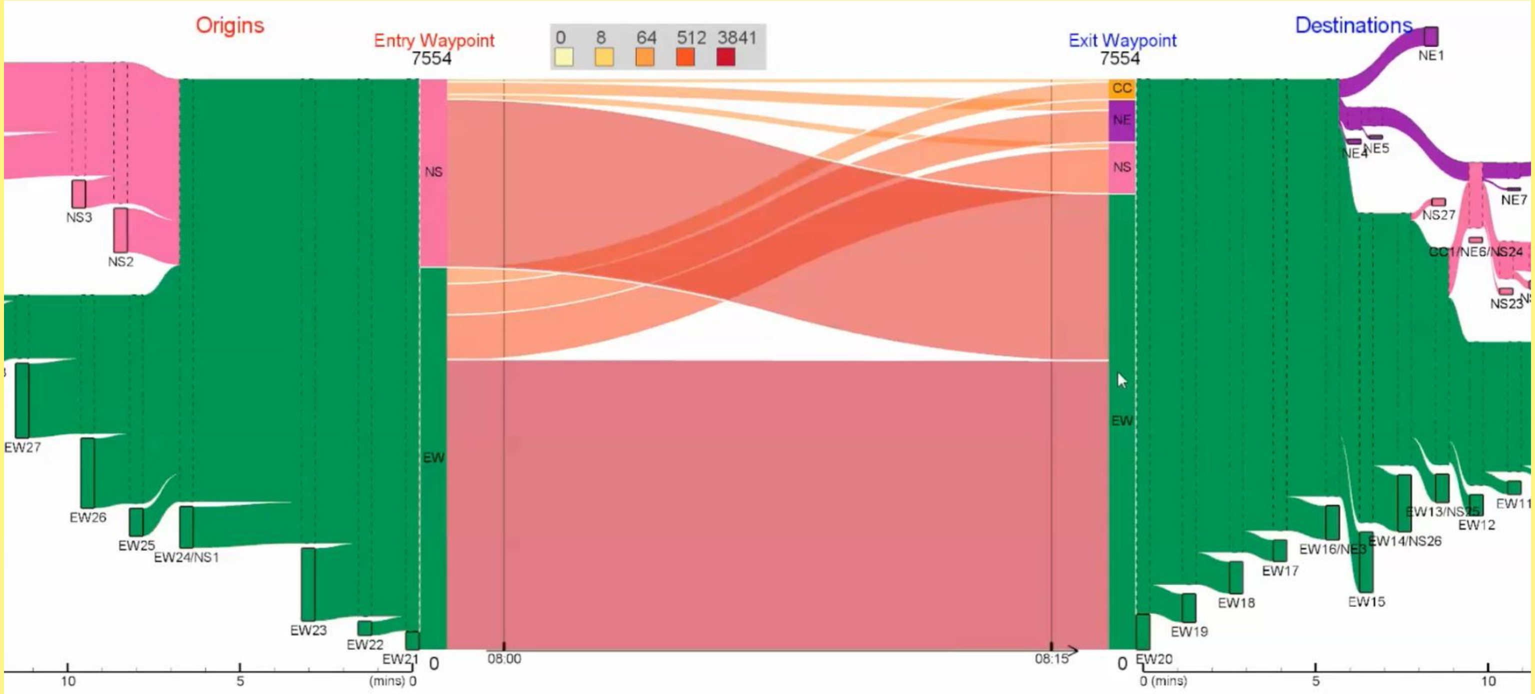
grouped origins



destinations

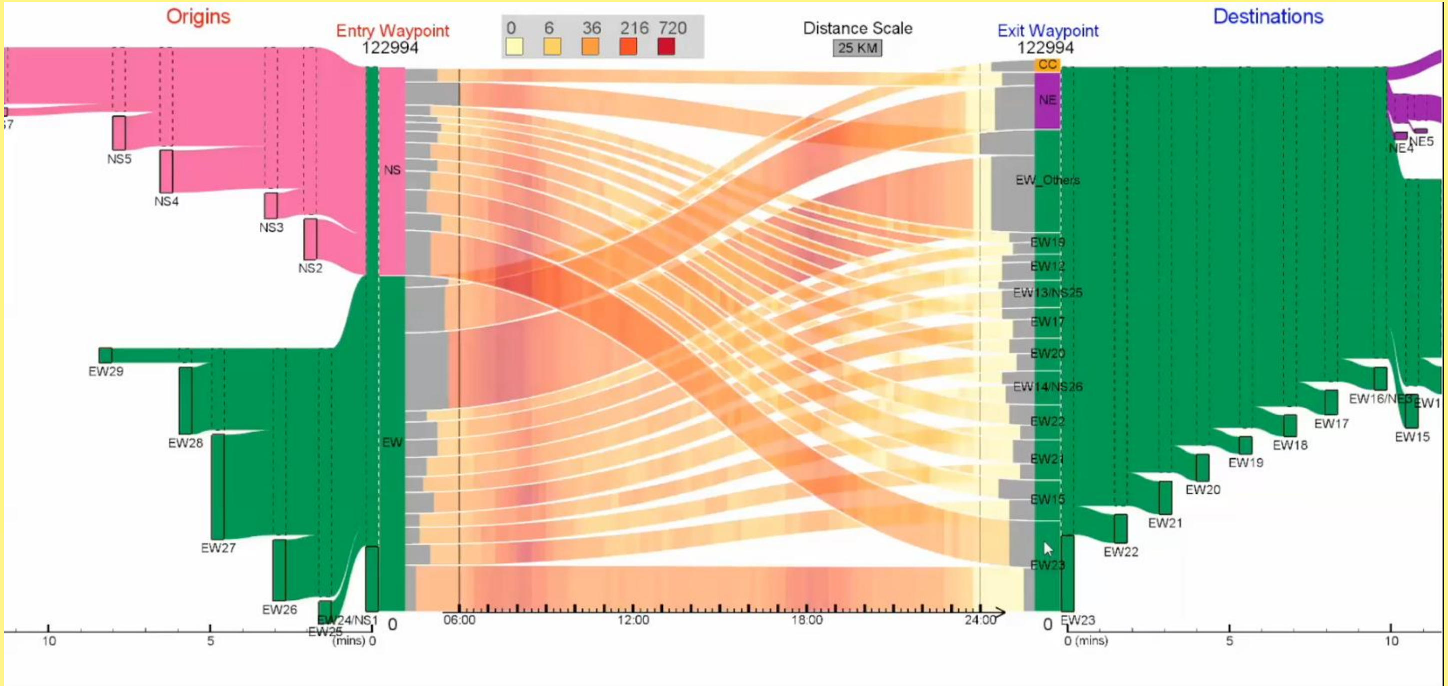
3.4.3 Design Considerations

- **Overview + Detail**: overview the OD flows by aggregating the origin/destination boxes, or explore details by decomposing them.



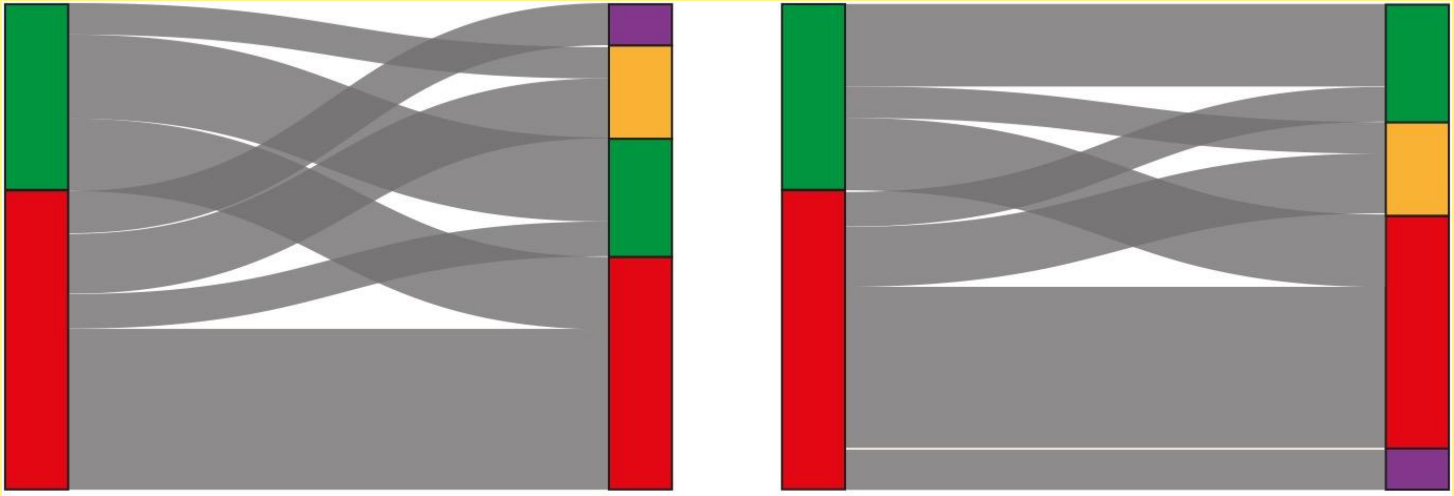
3.4.3 Design Considerations

- **Overlapping ribbons:** translucency to preserve visual connection and continuity, and opaque to highlight interested ribbons.



3.4.3 Design Considerations

- **Minimizing ribbon crossings:** reorder the origin and destination boxes, such that to minimize the overlapping area among the ribbons.
  1. Order the origin and destination boxes in ascending order of flow volumes
  2. Shuffle the boxes until the total overlapping area cannot be reduced
  3. Select the solution with least total overlapping area



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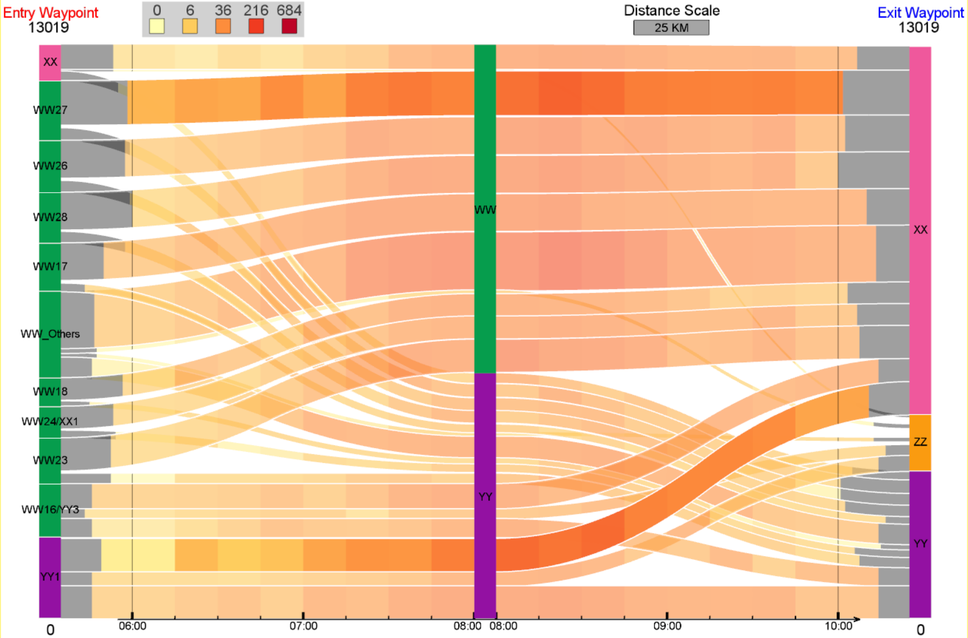
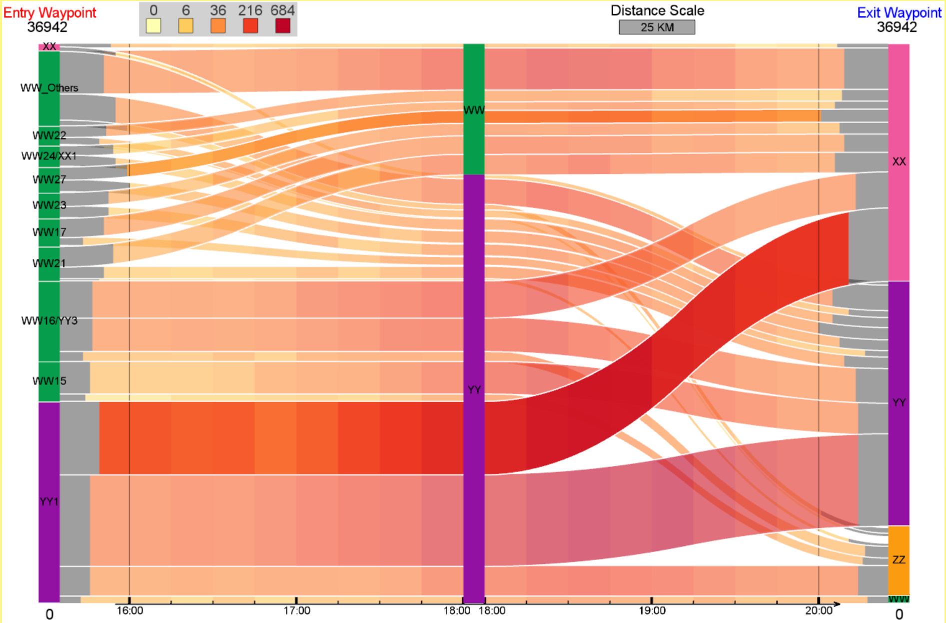
4.1 Transportation Network Usage Analysis

- Specify a pair of waypoints shown on the left.
- Explore the OD-flow temporal view in the morning (06:00 – 10:00) as shown on the right.



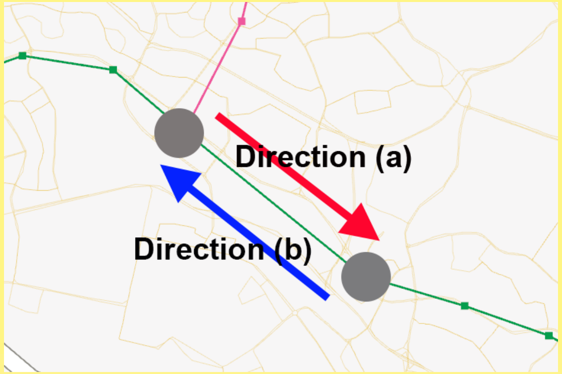
4.1 Transportation Network Usage Analysis

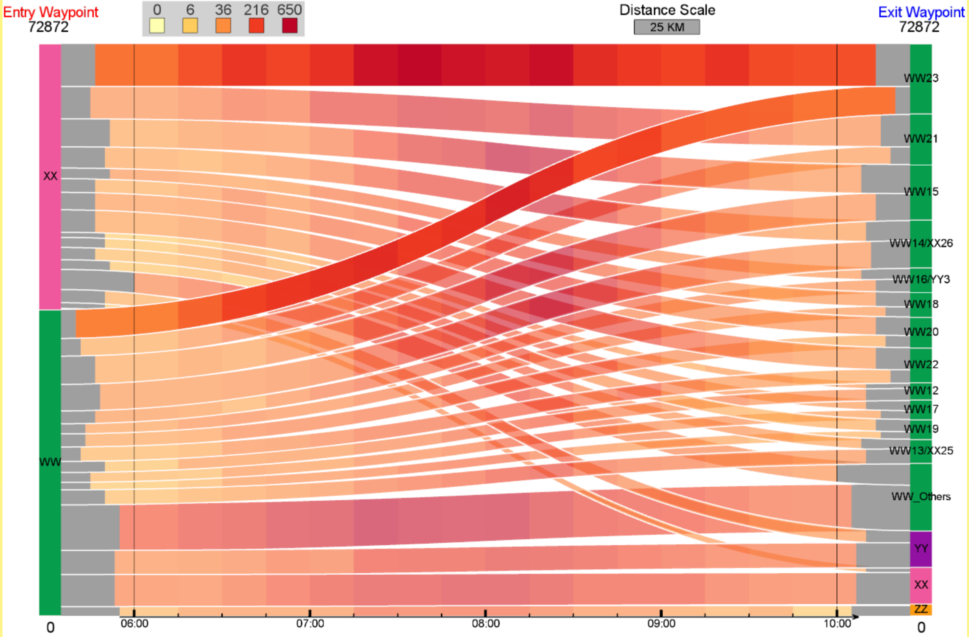
- Specify the same pair of waypoints.
- Explore the OD-flow temporal view in the afternoon (16:00 – 20:00) as shown on the right.



4.2 Daily Pendulum Movement Exploration

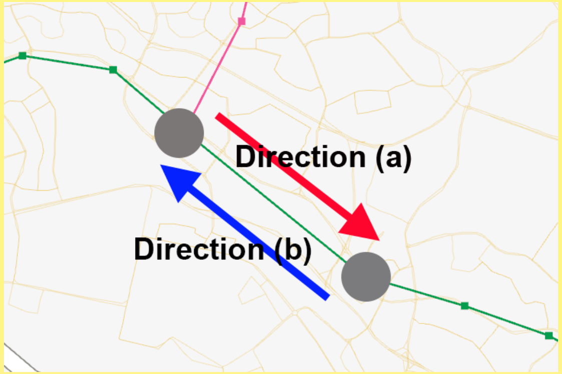
- Specify a pair of waypoints shown as the red arrow on the left.
- Explore the OD-flow temporal view in the morning (06:00 – 10:00) as shown on the right.

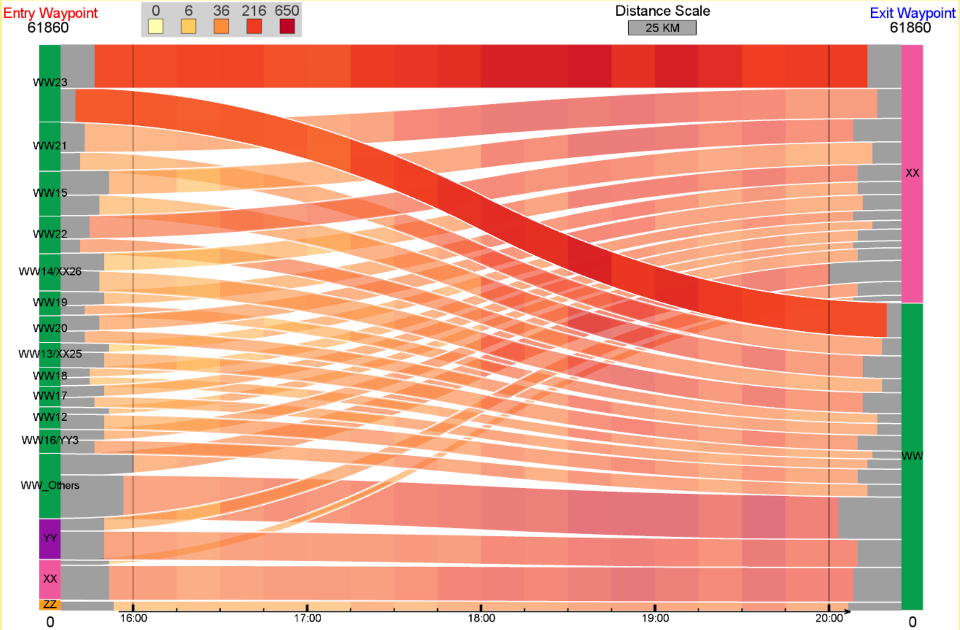




4.2 Daily Pendulum Movement Exploration

- Specify a pair of waypoints shown as the blue arrow on the left.
- Explore the OD-flow temporal view in the afternoon (16:00 – 20:00) as shown on the right.





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5.1 Conclusion

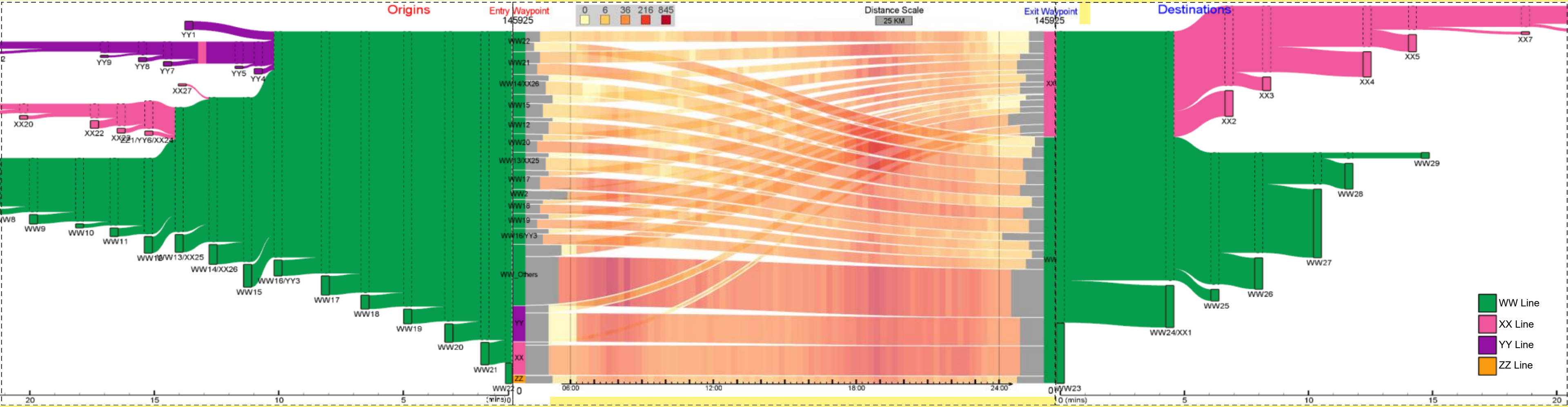
- We explore *waypoints-constrained OD patterns*
  - We introduce the concept of *waypoints-constrained OD patterns,* and develop three waypoints specification modes and relevant trips query mechanism.
  - We design *waypoints-constrained OD view* integrated with three visualization components: *In-flow view*, *Out-flow view*, and *OD-flow temporal view*.
  - We explore two case studies with transportation researchers, and present their feedbacks on the interface design.

5.2 Future Work

- We plan to integrate more visual elements, e.g., detailed distance and velocity views.
- We would like to explore other forms of origins and destinations grouping, such as by administrative regions.
- We plan to explore multiple pieces of transportation data over years, and analyze calendar patterns in the OD flows to explore the effect of recent upgrades in the transport system.

Thank You!

Zeng Wei (zeng@arch.ethz.ch)



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